MIL-T-29019(TD)
16 October 1970
SUPERSEDING
335-324
17 April 1969 and
3313-508
9 April 1967

MILITARY SPECIFICATION

TRAINING AID, DEMONSTRATOR: AUTOMOTIVE EQUIPMENT GENERAL SPECIFICATION FOR

This specification has been approved by the Naval Training Device Center, Department of the Navy

1. SCOPE

1.1 This specification covers the general requirements for the production of a panel-mounted, enlarged scale sectionalized mockup of an automotive equipment demonstrator.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bid or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-C-320	Chromium Plating (Electrodeposited)
QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-corner

Military

MIL-P-116	Preservation; Methods of
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-L-10547	Liners, Case and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible

FSC 6910

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MIL-T-23991	Training Devices, Military; General Specification for

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-171	Finishing of Metal and Wood Surfaces

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Materials. Except as specified in 3.1.1 through 3.1.5, materials shall be as specified in the detail specification.

3.1.1 Wood.- Wood used shall be select grades of birch, poplar or maple having good appearance and finishing qualities, suitable for either natural or paint finishes. Wood shall be properly seasoned, stablized, free from checks, splits, crossgrain and seasoning defects which may cause warping.

3.1.1.1 Plywood.- Plywood shall have a minimum of five plies or have a solid wood core. Outer surfaces shall be hardwood.

3.1.2 Metals.- Selection of metal parts shall be limited to the corrosion-resistant materials listed under corrosion-resistant metals of MIL-T-23991 or be of a material given a corrosion-resistant treatment or coating.

3.1.2.1 Aluminum.- Where the use of lightweight metal is required, aluminum alloy conforming to the aluminum requirements of MIL-T-23991 shall be used. (See 3.7.2.1).

3.1.3 Plastics.- Plastics used shall be durable, resistant to cracking, brittleness, discoloration, deformation or other defects upon aging or when subjected to the environmental conditions of 3.3. Transparent plastic components shall be acrylic.

3.1.4 Adhesives.- Adhesive materials shall be selected to meet the requirements of 3.1.4.1 and 3.1.4.2.

3.1.4.1 Wood glue.- Wood glue shall be a type not adversely affected by moisture, paint or lacquer.

3.1.4.2 Cements.- Cements for joining plastics shall be those cements recommended by the manufacturer of the plastic and tested by the supplier as suitable for strong homogeneous bonds that will not be affected by the environmental conditions of 3.3.

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3.1.5 Paints.- Paints used on metals shall be satin surface baked-on enamels or be equivalent in color fastness, scratch resistance, elasticity and bonding under the environmental conditions of 3.3. Wood and plastic components to be painted shall be sealed, primed and then finished with satin-surface lacquer. All sealers and primers used with paints shall be those recommended by the manufacturer of the paint.

3.2 Design.- The training aid shall consist of an enlarged sectionalized mockup of an item of automotive equipment mounted on a free-standing panel. The training aid shall be designed in accordance with the design requirements of the detail specification to illustrate component arrangement, function and theory of operation of the equipment represented.

3.3 Reliability.- The training aid shall be capable of withstanding handling by students for a period of ten hours per week for three years and shall not be adversely affected in any manner during handling, storage and transportation under ambient conditions of temperature ranging from -10° to +150° Fahrenheit, and of relative humidity up to 95 percent. The training aid shall be constructed so that no fixed part, subassemblies or assemblies shall become loosened or free under handling and storage conditions and no movable part shall bind or seize.

3.4 Construction.- The training aid shall be constructed as specified in the detail specification. Construction methods or materials may be varied in order to permit more practical and economical manufacturing processes and tooling practices. The changes will be acceptable provided these changes do not adversely affect the training value or durability of the training aid. All requests for variances shall be in the form of preliminary drawings or data, submitted for written approval by the Contracting Officer, prior to the fabrication of any component affected by the variance.

3.5 Details of components.- The details of components shall be as specified in the detail specification.

3.6 Dimensions and tolerances.- Dimensions and tolerances shall be as specified in the detail specification. Dimensions and tolerances not specified in the detail specification shall be consistent with functional design, inherent qualities of the material being used, and shall afford freedom from binding in mechanical fits.

3.7 Finish and color.- Except as specified in 3.7.1 through 3.7.2.2.2, finish and color shall be as specified in the detail specification.

3.7.1 Painted surfaces.- All surfaces of the training aid to be painted shall be prepared for painting and painted in accordance with the organic finishes requirement of MIL-STD-171.

3.7.2 Unpainted surfaces.- All unpainted metal surfaces of the training aid, except aluminum and corrosion-resistant steels, shall be plated.

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3.7.2.1 Aluminum.- Aluminum shall be anodized in accordance with the requirements for Type II, Class 2 coatings of MIL-A-8625.

3.7.2.2 Corrosion susceptible metals.- Metals requiring application of plating shall be plated in accordance with QQ-C-320 or QQ-C-416 as applicable.

3.7.2.2.1 Cadmium plating.- Where required, cadimum plating shall be Type I, Class 2 in accordance with QQ-P-416.

3.7.2.2.2 Chromimum plating.- Where required, chromium plating shall be Type II, Class 1, in accordance with QQ-C-320.

3.8 Nameplates or product markings.- Nameplates or product markings shall be as specified in the detail specification.

3.9 Nomenclature plates.- Nomenclature plates shall be provided for identification of the training aid, discs for identification of individual components or areas, and legend plates for listing the components or areas represented by the discs. Nomenclature plates shall be engraved in plastic, brass or aluminum and shall be permanently affixed to the training aid.

3.9.1 Identification plate.- The identification plate shall have the applicable title of the training aid in letters 1 1/2 inches high on a plate 2 inches high, by appropriate length. The plate shall have a background color identical with the color of the surface to which it is mounted and the letters shall be filled with black or white to provide a strong contrast with the background color.

3.9.2 Discs.- Discs for identification of components or areas, as listed in the detail specification, shall consist of a letter of the alphabet beginning with letter "A", 1/2 inch high on a disc 7/8 inch in diameter. Each disc shall have a background color identical to the color of the component or area represented. The alphabetical letter shall be filled with black or white to provide a strong contrast with the background color. The discs shall be permanently affixed to the component or area represented. Where it is clearly impractical, due to the lack of space or peculiar characteristics of the material, to affix the disc to the component represented, the disc shall be affixed to an area of the background panel adjoining the component represented. Where the background panel area does not adjoin the component represented, the disc shall be permanently affixed to an adjoining component. Discs not affixed to the component represented shall include a pointer to clearly indicate the component represented and the apex of the pointer shall visually appear to adjoin the component.

3.9.3 Legend plates.- The legend plate shall be of suitable size to provide space for listing the nomenclature and identifying letters of all components or areas as listed in the detail specification. The markings shall have letters 1/2 inch nigh. Lines of lettering for identifying each component shall be 1/4 inch apart. A space of 3/4 inch shall separate

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the listing of components. The legend plate shall have a background color identical with the color of the surface to which it is mounted. The identifying letters, corresponding to the discs of 3.9.2, shall be filled with black or white to provide a strong contrast with the background color. The nomenclature shall be filled with the identical color of the component or area represented.

3.10 Workmanship.- The training aid shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness in the machining and fabrication of components and assemblies. There shall be no burrs or sharp edges. Scribed lines, where required, shall be uniform and neat and no loss of sharpness of detail shall be apparent. All finished surfaces shall be free from toolmarks, scratches and blemishes of all kinds.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspections to be performed are classified as follows:

(a) In-process inspection

(b) Quality conformance inspection.

4.3 In-process inspection.- In-process inspection shall include such visual and mechanical testing of materials, subassemblies, parts and accessories (including purchased standard parts) during the manufacturing process as may be necessary to assure conformance to all the requirements of this specification.

4.4 Quality conformance inspection.- Quality conformance inspection shall be as specified in 4.4.1 through 4.5.

4.4.1 Preproduction inspection.- Preproduction inspection shall be conducted on a preproduction sample and shall consist of all the examinations specified in 4.5.

4.4.2 Production inspection.- Production units shall be inspected to the extent necessary to maintain quality control and conformance to the Government-accepted preproduction sample.

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4.5 Examinations.- Examinations shall be performed to verify conformance to the following requirements:

(a) Materials

(b) Construction

(c) Details of components

(d) Dimensions

(e) Finish and color

(f) Nameplates or product markings

(g) Nomenclature plates

(h) Workmanship

(i) Preparation for delivery.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be Level A or C as specified by the procuring agency. (See 6.2).

5.1.1 Level A.- For Level A, each device shall be preserved and packaged for mechanical and physical protection in accordance with MIL-P-116, Method III.

5.1.2 Level C.- For Level C, preservation and packaging shall conform to the manufacturer's commercial practice.

5.2 Packing.- Packing shall be Level A or C as specified by the procuring agency. (See 6.2).

5.2.1 Level A.- Units packaged in accordance with 5.1.1 shall be packed in exterior containers conforming to PPP-B-621, Class 2, or PPP-B-601, Styles A or B with steel banding. The gross weight shall not exceed the weight limitations of the shipping container. Each exterior shipping container shall be lined with a waterproof moisture barrier sealed with adhesive or tape to affect water resistance in accordance with MIL-L-10547. A case liner shall not be required where the package(s) and package dunnage have been treated to resist water.

5.2.2 Level C.- Units packaged as specified in 5.1.2 shall be packed in containers which will ensure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with common carrier rules

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and regulations as applicable to the mode of transportation and as may be the supplier's commercial practice when such meets the requirements of this level.

5.3 Physical protection.- All material shall be packaged and packed in containers constructed to provide a compact and nonshifting load. Adequate cushioning, blocking and bracing shall be provided within shipping containers in a manner designed to prevent damage due to shock, vibration or impact during shipment and storage.

5.3.1 Cushioning materials used shall be compatible with all finishes so that migrating solvents in cements, cushion or component will not adversely affect any part.

5.4 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The training aid, fabricated in accordance with this specification, is intended for use in classrooms to assist military personnel in understanding the configuration, relative size and locations of major components of automotive equipment.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) Title, number and date of the applicable detail specification

(c) Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2).

6.3 Definitions.- Definitions will be furnished upon written request to the procuring agency.

CUSTODIAN:
Navy-TD

PREPARING ACTIVITY:
Navy-TD
Project No. 6910-N255

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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